AI Fraud Detection for WooCommerce

User Manual v2.2.1

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Introduction

What is AI Fraud Detection?

AI Fraud Detection for WooCommerce is an intelligent plugin that protects your online store from fraudulent transactions using advanced machine learning. It analyzes each order in real-time and provides fraud risk assessment with explainable AI insights.

Key Benefits

✅ **Real-time Protection** - Instant fraud detection on every order
✅ **Automated Response** - Automatically hold suspicious orders
✅ **Bulk Processing** - Check hundreds of transactions via CSV upload
✅ **Explainable AI** - Understand why orders are flagged
✅ **Email Alerts** - Get notified immediately of fraud attempts
✅ **Zero Configuration** - Works out of the box with sensible defaults

How It Works

1. Customer places an order on your WooCommerce store
2. Plugin sends order data to AI fraud detection API
3. AI model analyzes 12+ risk factors in milliseconds
4. Plugin receives verdict: **Legitimate** or **Fraud**
5. Automated actions taken (hold order, send alerts)

6. Merchant reviews results and makes final decision

Quick Start

⚡ 3-Minute Setup

Step 1: Install Plugin (1 minute)

1. Download `wc-fraud-detection-v2.2.1-FIXED.zip`
2. WordPress Admin → Plugins → Add New → Upload Plugin
3. Install and Activate

Step 2: Configure (1 minute)

1. Go to WooCommerce → Fraud Detection
2. Enter API Endpoint: `http://localhost:8000`
3. Set Threshold: `0.7`
4. Enable all checkboxes
5. Test Connection → Save

Step 3: Test (1 minute)

1. Go to WooCommerce → Orders
2. Click any order
3. Click "Check for Fraud" button
4. View results!

Installation

System Requirements

Minimum:

- WordPress 5.8+
- WooCommerce 8.0+
- PHP 7.4+
- Active internet connection

Recommended:

- WordPress 6.0+
- WooCommerce 8.5+
- PHP 8.0+
- SSL certificate (HTTPS)

Installation Steps

Step 1: Download Plugin

Download the latest version: `wc-fraud-detection-v2.2.1-FIXED.zip`

Step 2: Upload to WordPress

1. Log into WordPress Admin Dashboard
2. Navigate to **Plugins** → **Add New**
3. Click **"Upload Plugin"** button
4. Click **"Choose File"** and select the ZIP file
5. Click **"Install Now"**

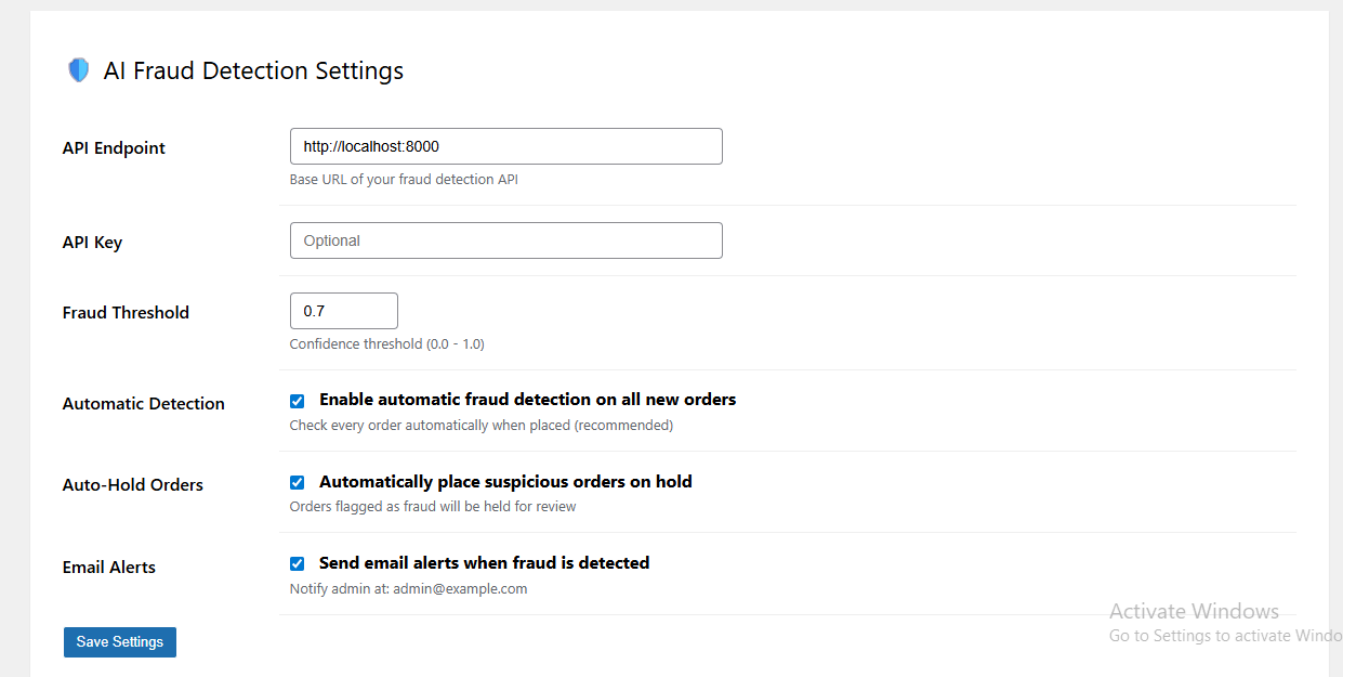
Step 3: Activate Plugin

1. Click **"Activate Plugin"** after installation
2. Look for new menu items under **WooCommerce**:
    - Fraud Detection (Settings)
    - Bulk Check (CSV)

Configuration

Settings Page

Access: **WooCommerce** → **Fraud Detection**



API Endpoint Configuration

What it is: The URL where your fraud detection API is running

How to configure:

1. Enter the API endpoint URL
2. For local development: `http://localhost:8000`
3. For production: `https://your-api-domain.com`

Important: Must include `http://` or `https://`

Fraud Threshold

What it is: Minimum confidence level to flag as fraud

How to set:

- Range: 0.0 to 1.0 (0% to 100%)
- Recommended: **0.7** (70% confidence)

Adjustment guide:

- **0.5-0.6:** Very sensitive (more alerts)
- **0.7:** Balanced (recommended)
- **0.8-0.9:** Conservative (fewer alerts)

Automation Settings

Automatic Detection

- ✅ **Enable** - Check all new orders automatically (recommended)
- ❌ **Disable** - Manual checking only

Auto-Hold Orders

- ✅ **Enable** - Automatically hold suspicious orders (recommended)
- ❌ **Disable** - Flag only, no status change

Email Alerts

- ✅ **Enable** - Send fraud notifications to admin (recommended)
- ❌ **Disable** - Silent operation

Testing Connection

1. Click **"Test Connection"** button
2. Wait 5-10 seconds
3. Look for success message:
    - ✅ Connection successful!
    - Status: healthy

    - Model loaded: Yes

If connection fails:

- Verify API is running
- Check endpoint URL is correct
- Ensure firewall allows outbound connections

Save Settings

1. Click **"Save Settings"** button
2. Confirm success message appears
3. Settings are now active!

Daily Usage

Automatic Fraud Detection

When automatic detection is enabled, the plugin works silently:

What happens automatically:

1. Customer places order
2. Plugin checks for fraud
3. If fraud detected:
    - Order status → **On Hold**
    - Email sent to admin
    - Order note added

No action needed - Plugin works in background!

Manual Order Checking

When to Use

- Review suspicious orders manually
- Re-check historical orders
- Verify before fulfillment

How to Check an Order

Step 1: Open Order

1. Go to WooCommerce → Orders
2. Click on any order

Step 2: Find Fraud Detection Box

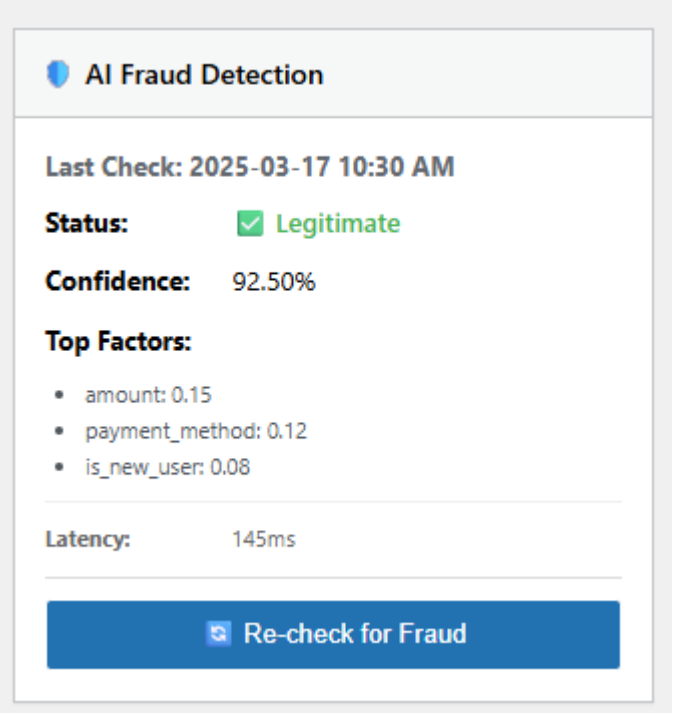
Look for **" 🛡️ AI Fraud Detection"** box in right sidebar

Step 3: Check for Fraud

Click "🔍 **Check for Fraud**" button

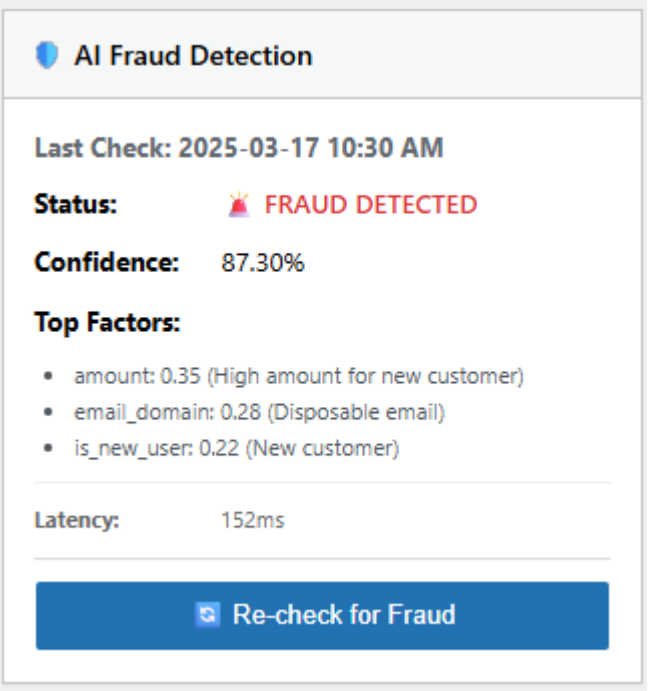
Step 4: View Results

For Legitimate Orders:



- Status: ✅ Legitimate (green)
- Confidence: 90%+ typical
- Top Factors: Low-risk indicators
- Latency: Processing time

For Fraudulent Orders:



- Status: 🚨 FRAUD DETECTED (red)
- Confidence: 70-90%+ typical
- Top Factors: High-risk indicators explained
- Latency: Processing time

Understanding Results

Confidence Score

What it means:

- 50-60%: Low confidence (borderline)
- 60-75%: Moderate confidence
- 75-90%: High confidence
- 90-100%: Very high confidence

Example:

- 92.5% legitimate = 92.5% certain order is safe
- 87.3% fraud = 87.3% certain order is fraudulent

Top Risk Factors

Common factors explained:

Factor	Meaning	High Risk When
`amount`	Transaction amount	Very high or very low
`is_new_user`	New customer	Yes + high amount
`email_domain`	Email provider	Disposable/temporary
`payment_method`	How they paid	Credit card (vs PayPal)
`billing_shipping_match`	Addresses match	No (different addresses)
`hour_of_day`	Purchase time	Late night (2-5 AM)

Example interpretation:

```
Top Factors:
• amount: 0.35 (Very high transaction)
• email_domain: 0.28 (Disposable email)
• is_new_user: 0.22 (New customer)
```

Translation: Order flagged due to large amount from new customer using temporary email.

Bulk CSV Upload

When to Use

- Analyze historical transactions
- Check multiple orders at once
- Generate fraud reports

CSV Format Requirements

Your CSV must have these columns:

```
order_id,amount,payment_method,customer_email,is_new_customer,billing_city,items_count
```

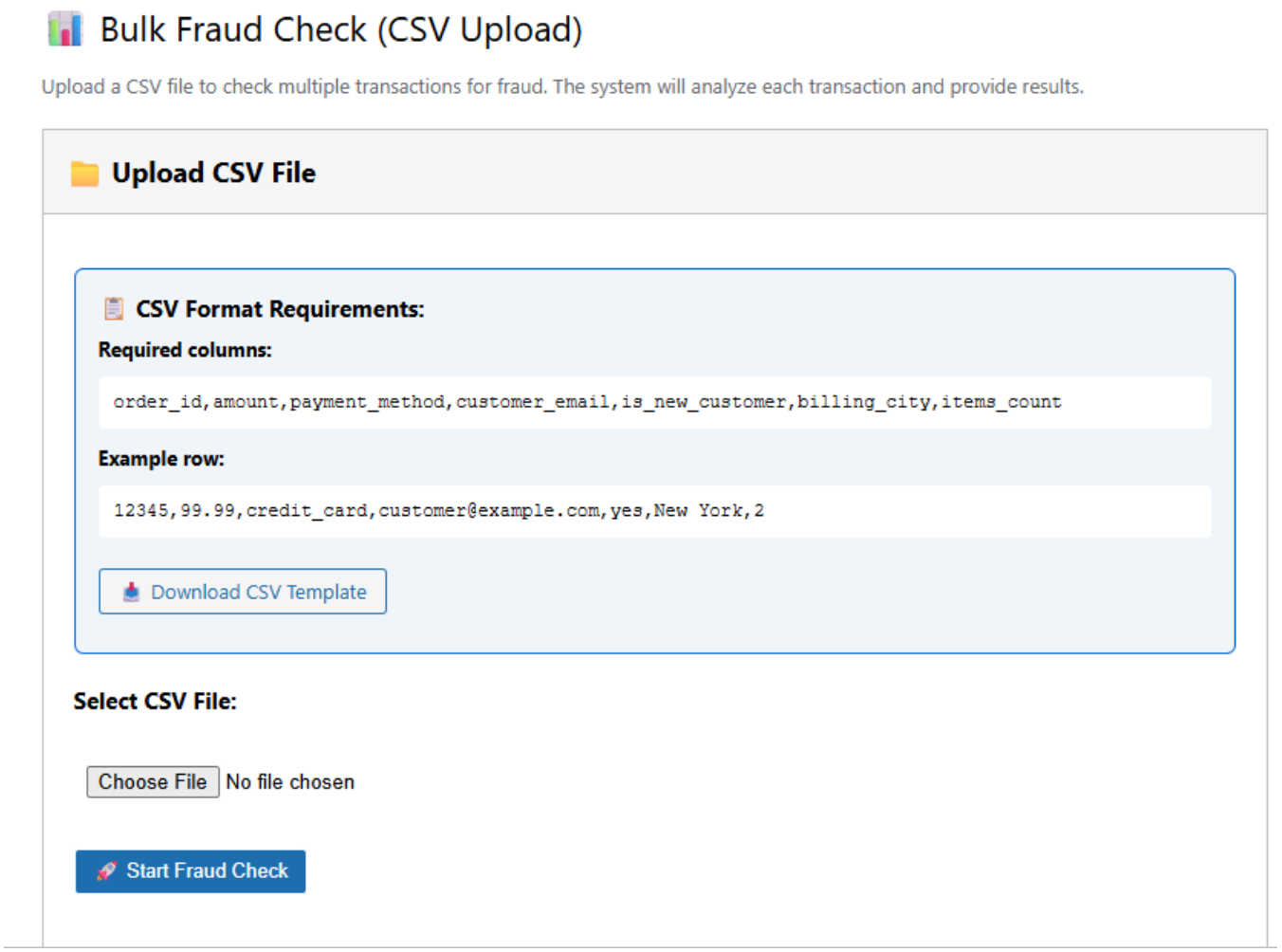
Example:

```
order_id,amount,payment_method,customer_email,is_new_customer,billing_city,items_count
12345,99.99,credit_card,customer@example.com,yes,New York,2
12346,149.50,paypal,john@test.com,no,Los Angeles,1
```

Upload Steps

Step 1: Access Bulk Check Page

Go to **WooCommerce** → **Bulk Check (CSV)**



Step 2: Prepare CSV File

Download template or create your own following format above.

Step 3: Upload File

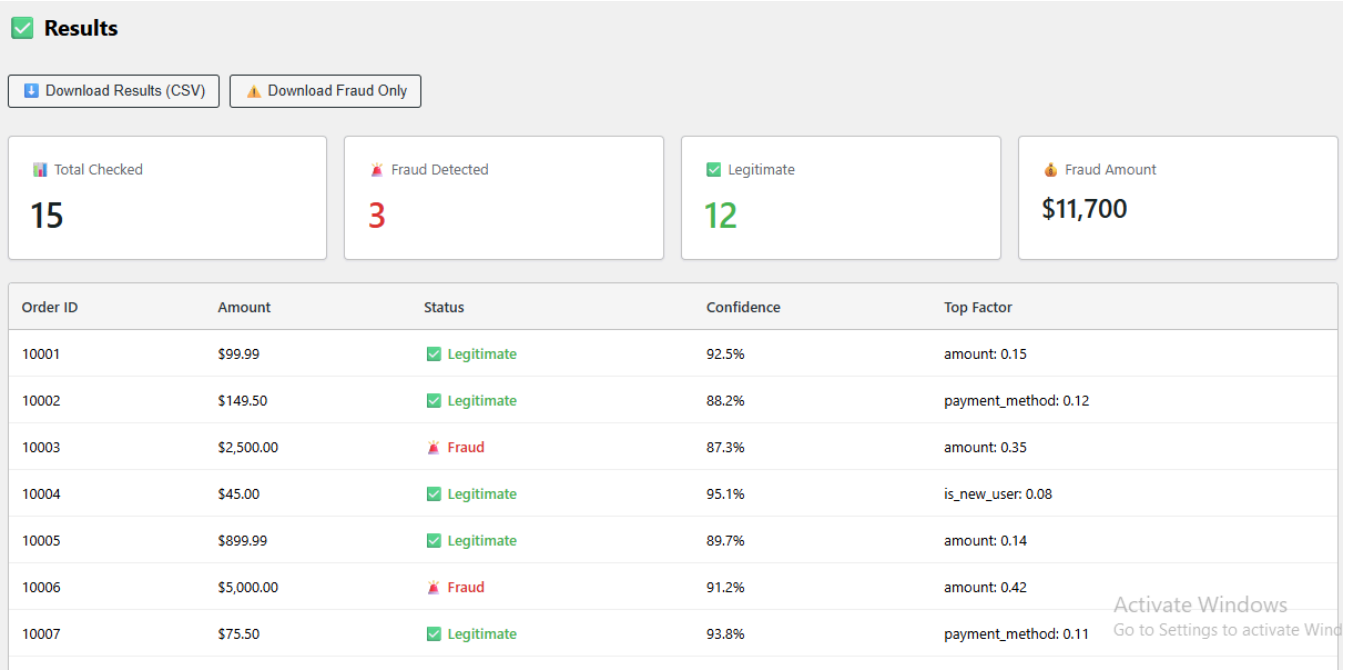
1. Click **"Choose File"**
2. Select your CSV file
3. Click **"🚀 Start Fraud Check"**

Step 4: Monitor Progress

Watch real-time progress:

- Progress bar shows percentage
- Status shows current batch
- Estimated time remaining

Step 5: View Results



Summary Cards:

- 📊 Total Checked
- 🚨 Fraud Detected
- ✅ Legitimate
- 💰 Fraud Amount

Results Table:

- All transactions with verdict
- Confidence scores
- Top risk factors

Step 6: Download Results

1. **Download Results (CSV)** - All transactions
2. **Download Fraud Only** - Just flagged transactions

Troubleshooting

Common Issues

❌ Connection Failed

Problem: Can't connect to API

Solutions:

1. Verify API is running at endpoint
2. Check firewall allows connections
3. Ensure endpoint URL is correct

4. Test in browser: `http://your-api/api/v1/health`

❌ Orders Not Checked Automatically

Problem: Auto-detection not working

Solutions:

1. Enable "Automatic Detection" in settings
2. Verify API connection works
3. Check WordPress error logs
4. Try manual check first

❌ CSV Upload Invalid

Problem: CSV file won't upload

Solutions:

1. Ensure proper headers (see format above)
2. Save CSV with UTF-8 encoding
3. Use provided template
4. Check file has line breaks

❌ Email Alerts Not Received

Problem: Not getting fraud alerts

Solutions:

1. Enable "Email Alerts" in settings
2. Check spam/junk folder
3. Verify admin email is correct
4. Test WordPress email functionality

FAQ

General Questions

Q: Is this plugin free?
A: Yes, MIT licensed and open source. You host the API yourself.

Q: Will it slow down checkout?
A: No! Detection runs after checkout completes, no customer delay.

Q: How accurate is it?
A: 90%+ recall, 85%+ precision on test data.

Setup Questions

Q: What if API is down?
A: Plugin logs error but doesn't block orders. Set up monitoring.

Q: Can I adjust sensitivity?
A: Yes! Change fraud threshold in settings (0.0-1.0).

Q: Does it work with PayPal?
A: Yes! Works with all WooCommerce payment methods.

Privacy Questions

Q: What data is sent to API?
A: Order amount, payment method, email domain (not full email), customer type, location, items count.

Q: What is NOT sent?
A: Credit card numbers, CVV, passwords, full emails, personal messages.

Q: Is it GDPR compliant?
A: Yes, when configured properly. API processes data temporarily only.

Support

Getting Help

📧 **Email Support**
tanveer030402@gmail.com

💬 **GitHub Issues**
https://github.com/tanveer-ahmed986/fraud_detection_system_ecommerce/issues

📚 **Documentation**
See GitHub repository for full technical docs

Before Contacting Support

Please have ready:

1. Plugin version (2.2.1)
2. WordPress version
3. WooCommerce version
4. PHP version
5. Error messages or screenshots
6. Steps to reproduce issue

Appendix

Performance Metrics

Speed:

- Prediction: 50-200ms typical
- CSV: 10 transactions per batch
- UI: Real-time updates

Accuracy (Test Data):

- Recall: 90%+ (catches fraud)
- Precision: 85%+ (accuracy)
- False Positive Rate: <5%

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Credits

Developer: Tanveer Ahmed
Framework: WordPress, WooCommerce, FastAPI
ML Model: Custom-trained fraud detection
License: MIT (Open Source)

Thank you for using AI Fraud Detection for WooCommerce!

Version: 2.2.1
Release Date: March 17, 2026
Status: ✅ Production Ready

For support, visit: https://github.com/tanveer-ahmed986/fraud_detection_system_ecommerce

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End of User Manual v2.2.1